

23-Apr-2025

Dear Mr Freedman:

I am writing to inform you that your manuscript RSPB-2025-0360 entitled "A unifying theoretical framework for tick-borne disease risk to explain conflicting results of enclosure experiments across scales" has, in its current form, been rejected for publication in Proceedings B.

This action has been taken on the advice of referees, who have recommended that substantial revisions are necessary. With this in mind we would be happy to consider a resubmission, provided the comments of the referees are fully addressed. However please note that this is not a provisional acceptance.

The resubmission will be treated as a new manuscript. However, we will approach the same reviewers if they are available and it is deemed appropriate to do so by the Editor. Please note that resubmissions must be submitted within **six months** of the date of this email. In exceptional circumstances, extensions may be possible if agreed with the Editorial Office. Manuscripts submitted after this date will be automatically rejected.

Please find below the comments made by the referees, not including confidential reports to the Editor, which I hope you will find useful. If you do choose to resubmit your manuscript, please upload the following:

- 1) An **anonymised** 'response to referees' document including details of how you have responded to the comments, and the adjustments you have made.
- 2) A clean copy of the **anonymised** manuscript and one with 'tracked changes' showing how you addressed the referee and editor comments.
- 3) Line numbers in your main document.
- 4) Data - please see our policies on data sharing to ensure that you are complying (<https://royalsociety.org/journals/authors/author-guidelines/#data>).

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Sincerely,

Professor Hans Heesterbeek
mailto:proceedingsb@royalsociety.org

Associate Editor
Comments to Author:

The reviewers must pay careful attention to these two reviews, particularly Reviewer 2. As hinted at by that reviewer, usage of concepts like "dilution" and "amplification" are inconsistently (sometimes sloppily) used in the literature. The authors must define their use of these terms precisely and justify that usage where it differs from more narrow, technical definitions -- or simply use other language altogether.

This is a great point, thank you, we totally agree. We have now included more rigorous definitions of "dilution" and "amplification" in the main text and clarified our discussions of these concepts as relates to our model in various places. We cover these changes in more detail in the following responses.

Reviewer(s)' Comments to Author:

Referee: 1

Comments to the Author(s)

It is gratifying to see a manuscript that puts the concept outlined in Buck and Perkins 2018 into theoretical terms. It's even more gratifying to see that the mathematical model confirms our strong suspicions that Titcomb et al.'s results would reverse, had they used larger enclosure plots.

Thank you very much, we are very grateful that you enjoyed our unifying theoretical model.

Below I detail a few minor suggestions, along with a suggestion to simplify the model (or at least clarify to readers) that TBP risk changes with enclosure size because tick density depends on enclosure size, not because infection prevalence depends on enclosure size.

This is a great point, we have now clarified that TBP risk changes with enclosure size because overall tick density depends on enclosure size, not because infection prevalence depends on enclosure size. We clarify this in both the Results (lines 161-163) and Discussion (lines 298-303), and confirm with a new Fig. S1 in the Supplement that infection prevalence is practically constant across space inside the enclosures (although it is about 10% higher inside the enclosures than outside).

Line 10: I don't understand the use of the word "incompetent" here. The large mammals that are excluded from enclosures (e.g., deer) are generally excellent hosts for adult ticks. Do you mean they're incompetent hosts for TBP? I think this word should be eliminated here because it's confusing. Note: after reading the rest of the paper, I understand what the authors mean here (explained in line 98), but the fact remains that it's still highly confusing in the abstract because deer amplify tick populations, at least over large spatial and temporal scales.

Thank you for bringing this up, we agree that we need to be more specific on this. In the Introduction (line 41) we have clarified that large mammals are incompetent “with respect to TBP transmission”, and in the Abstract where we are more word-limited we have changed “competent” and “incompetent” to “transmission-competent” and “transmission-incompetent”.

Figure 1 caption: The ticks are modeled as susceptible, exposed but not yet infectious, infectious, and recovered. But do ticks really “recover” from TBP’s? Is this documented? I note that in the text (line 96), there’s no “recovered” class.

Indeed, ticks do not recover from infection in our model, but the hosts in our model can recover, following previous modeling work. We don’t believe we have mentioned anywhere that ticks can recover in our model, but please let us know if this is incorrect and we will fix it. We have also clarified in our Model overview (line 105) that “infectious ticks remain infectious for the rest of their lives”. We also clarify this in the caption for Fig. 1 (“and r for recovered and immune (hosts only)”).

Line 219: “We show that this scale dependence can result from a balance between a strong dilution effect at the exclosure’s edge, where the lack of deer increases infected tick densities, and a strong amplification effect far away from the exclosure’s edge, where the lack of deer decreases infected tick densities due to the limits of ticks’ rodent-based dispersal. This phenomenon, that dilution is dominant where deer have a strong influence while amplification is dominant where deer have a weak influence, is consistent with the notion from previous empirical and theoretical work [25, 27, 28] that deer are amplification hosts at low densities and dilution hosts at high densities.” This paragraph needs to be re-written to take into account exclosure size. If rodent-based dispersal is limited at the center of exclosures, then where did all those ticks come from? (Answer: they were imported by rodents, but then failed to find a large mammal host because large mammals are excluded.) Deer don’t have a “strong influence” or a “weak (not week) influence” in different places, they simply have different kinds of influences. In a small plot, questing ticks are distributed among available deer, such that there are a lot of questing ticks where deer are absent, and few questing ticks where they are present. In a large plot, deer produce ticks, such that there are few ticks where deer are absent, and a lot of ticks where they are present. This explanation is very different from the explanation for deer at low densities amplifying ticks and deer at high densities diluting ticks. I’d speculate that the later pattern occurs because when deer are rare, their density is what regulates tick populations, whereas when they are common, other factors (e.g., availability of other hosts or abiotic factors) regulate tick populations.

We appreciate this keen insight. You are correct to point out that we were hastily drawing connections between the effects of deer presence/absence that we observed in our model and the phenomenon observed in other works of deer being amplification/dilution hosts at different densities. We have removed the last sentence from this paragraph (first paragraph of Discussion) to avoid this ill-founded connection, and clarified in the previous sentence (lines

225-226) that the lack of ticks in the center of large exclosures is due to the limits of rodent-based dispersal from areas with deer to areas without deer.

Line 241: delete the word “infected” here. Ref 16 (Buck and Perkins 2018) did not test whether ticks were infected or not. Same thing in line 263, 267. This actually brings up a larger point, which is that based on my reading, there is no need to model TBP infection at all, because it’s not scale-dependent. What is scale-dependent is tick density. In other words, the results would not change if the TBP was eliminated from the system. For this reason, I think the manuscript would benefit from eliminating TBP’s from the model (which would make the model a lot simpler!), along with the methods and results section. Simply state that we’re assuming that some proportion of ticks (and hosts) are infected, and that TBP infection risk changes with exclosure size due to the fact that tick densities change with exclosure size. As it stands currently, a reader who isn’t reading very carefully could assume that TBP infection risk changes with exclosure size because tick (or host) infection rates change with exclosure size, which is not true. At a minimum, this needs to be made much clearer to readers. If you choose to eliminate TBP’s from the model/methods/results sections, TBP’s could still be retained in the intro and discussion sections to make applications clear to readers.

Thank you for catching this mistake, we have deleted the word “infected” in these spots as you mentioned. And we appreciate you bringing up the fact that it is overall tick density rather than infected tick density which is driving the observed spatial gradients, as this is a very important point to discuss. As mentioned before, we now clarify this fact in both the Results (lines 161-163) and Discussion (lines 298-303), and confirm with a new Fig. S1 in the Supplement that infection prevalence is relatively constant across space within our simulated exclosures. We believe it is still important to model and report on infected tick densities specifically, as these are the ticks of most direct public health concern, and we do find a uniformly higher prevalence inside the exclosures than outside (about 10% greater due to the dilution effect). Furthermore, it is an important insight (which we now discuss in lines 301-303) that the same results appear from looking at total tick densities vs. infected tick densities, as this implies that one can be used as a proxy for the other in experimental exclosure studies. We hope that our additional clarifications and discussions of these topics are sufficient as we keep our inclusion of the TBP in the model.

Referee: 2

Comments to the Author(s)

Please see attached file

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Introduction

Introduction (lines 28-35) of the dilution and amplification effect seems to be referring to the case of ungulates and Lyme disease transmission, this is a specific case, the general concept of dilution and amplification should be explained and then the case where a species can amplify the vector and both amplify and dilute pathogen transmission needs to be carefully explained. The use of dilution/amplification terminology throughout the manuscript is unclear and needs to be reworded. For example line 38 – removal of a host leading to increased DIN suggestive of dilution? Also removing larger hosts resulted in decreased DIN suggestive of amplification?

We appreciate you indicating this point of confusion. We have included a more general yet more precise definition of “dilution” and “amplification” in the Introduction in lines 28-31. Simply, we follow Rohr et al. 2020 in defining dilution to include any mechanism that produces a negative correlation between biodiversity and disease risk, and amplification to include any mechanism that produces a positive correlation between biodiversity and disease risk. As a result, amplification includes not only cases where adding large mammals increases DIN, but also cases where removing large mammals decreases DIN, since both represent a positive relationship between biodiversity and disease risk. Similarly, dilution also includes cases where removing large mammals increases DIN. We recognize this is unintuitive, so we have made this more clear in lines 30-31. We also now cite Gandy et al. 2022 to address the important

possibility that amplification and dilution are not mutually exclusive and may happen concurrently (lines 35-36).

Paragraph starting on line 36 does not include the results of all exclosure experiments including small exclosures which showed different results to those reported (eg Gandy et al 2021). This section should be updated to reflect all studies and a table added of the studies and main findings.

Thank you for catching this oversight. We have now included Gandy et al. 2021 in the Discussion (line 272-274) as a crucial counterexample to the pattern we seek to explain, and cite them at the end of the next paragraph (lines 293-294) for its discussion of ecological factors that may influence the results of exclosure experiments (specifically referencing their “ecological cascade”, which we call “release from herbivory”). We could not find any other exclosure studies of tick abundance that found similar contrary patterns, but please let us know if there are any others we are missing. As our paper is not a review or meta-analysis, we feel that including an exhaustive table of all exclosure studies would be beyond the scope of the paper; however, we have included references to four more studies (Stafford et al. 2003, Gilbert et al. 2012, Keesing et al. 2013, Titcomb et al. 2017) in lines 41-43 of the Introduction in order to be more complete. Again, please let us know if there are any other important relevant studies which we have overlooked.

Line 70 on, introduces concept of allowing ticks to move independently, not something I have seen any evidence for in the literature, ticks are constrained by finite energy resources and optimising activity to quest and maximise host seeking opportunities, and by saturation deficit and need to rehydrate. Vertical movement in the vegetation is described with little (<2m) horizontal movement for 3 host Ixodid ticks.

Thank you for pointing out our oversight in not citing our sources for including questing tick movement. We now cite 7 papers which all find some level of lateral movement of questing ticks, citing them in the Introduction (line 75), Methods (lines 121-122), and Results (line 178). This appears to be a greatly context- and species-dependent phenomenon, as these studies all found varying levels of questing tick movement (all the way up to 36m in one day, though this is surely an outlier) while other studies reported no questing tick movement at all. We opt to keep including questing tick movement in our paper, since it is evidently noted in the literature, and it is a phenomenon which has crucial implications for the qualitative behavior of our model. Even if not all questing tick species have significant lateral movement in all habitats, it is still important to cover this case. However, we have reduced the average daily tick movement from 5 m/day to 1 m/day to be more in line with the results found from these 7 sources.

Line 77, Tick sharing hypothesis – can a sentence of two be added so reader doesn’t have to go to the reference to find out what this is?

Of course, we are happy to make this more clear. This sentence (lines 81-84) has been rewritten and expanded to better define “tick sharing” in the manner of Perkins et al. 2006 and Buck and Perkins 2018, where the term originated. Please let us know if this is still unclear at all. We also expand on it in more detail where it is relevant to do so in the Discussion (lines 251-257).

Line 112 – Rodents suffer infection based mortality – what is the evidence for this from the literature?

The model from Pugliese and Rosà 2008 on which we base our model and parameter values includes rodent-based mortality, which is why we include it as well. They take their parameter estimate of rodent-based mortality by tick-borne encephalitis from Randolph et al 1996. We now include a reference to both of these papers in line 122 where we first bring up rodent infection-based mortality.

Of course, not all tick-borne pathogens may lead to mortality in rodent hosts. As a sensitivity check, we have now included Fig. S4 in the supplement which shows the model results without rodent infection-based mortality, producing qualitatively identical results. We discuss this now in the Results (lines 165-166) and the Discussion (lines 307-309).

Ditto ticks moving in Brownian motion in between feedings, this doesn't match up with biological reality of the study system.

Indeed, Brownian motions are far from being an accurate way of describing any animal movement. In reality, tick movement, as described by the papers on questing tick movement we cite (in lines 75, 121-122, and 178) is driven by many complex variables from microclimatic conditions to potential hosts' odors. Since we are modeling a generalized, species non-specific system, we use Brownian motion as a first approximation to macroscopic animal movement patterns. We use this null model with the hope of incorporating more specific movement patterns suited to particular species as an important consideration for future work. We now make these points and acknowledge the limitations of Brownian motion in the Intro (lines 96-98) and the Discussion (lines 284-288)

Results

Interesting results on interactions between enclosure size and rodent movement.

148- again mentions a strong dilution effect in relation to lack of deer and high DIN, this is not my understanding of this effect, this represents amplification ? And lack of ticks in the centre of the enclosure wrongly described as amplification?

Again we are sorry for this confusion. As we explained earlier and now explain better in the Introduction (lines 28-31): Dilution describes how adding new hosts decreases DIN, which directly implies that removing hosts increases DIN. Similarly, amplification describes how adding new hosts increases DIN, which directly implies that removing hosts decreases DIN. We

better explain this in the Introduction with our more general definition of dilution and amplification (adapted from Rohr et al. 2020), which is that dilution represents a negative correlation between host diversity and disease risk, and amplification represents a positive correlation between host diversity and disease risk.

Figure 4 – the different tick density scales make comparing the different sized enclosures more complicated, can this be changed to make it more intuitive?

We did think about this, but unfortunately if we make all of the plots in Fig. 4 on the same scale, then it becomes very difficult to see the differences in densities within each plot, since the range of densities with all plots combined is much greater than the ranges of each plot individually. We believe this is okay for the purposes of this figure, however, since this figure is emphasizing the qualitative nature of the spatial gradients in each plot, rather than comparing the magnitude of the densities between small and large enclosures. Furthermore, Fig. 2 already accomplishes this latter task, quantitatively comparing the densities for enclosures of all sizes.

Section 3.2 – What is the justification for ticks moving independently 5m a day? This doesn't match up with any tick biology for Ixodid three host ticks I know of. I don't think this section should be included.

As mentioned earlier, we now include citations to 7 sources that support independent moving from questing ticks (cited in lines 75, 121-122, and 178). However, we have lowered the average questing tick movement from 5 m/day to 1 m/day, as this is a more realistic estimate in line with the results from these sources. We believe it is important to include this section, since questing tick movement is supported from some cases in the literature, and it has crucial implications for the qualitative and analytic results of our model. Specifically, adding questing tick movement to the model produces reversed and non-monotonic tick density gradients that match those found in the literature (e.g. Perkins et al. 2006, Titcomb Perkins et al. 2017, Buck and Perkins 2018, as we discuss in lines 242-251). Thus, questing tick movement may be an important mechanism to consider when interpreting spatial density gradients found from enclosure experiments.

Discussion

First paragraph, again confusing references to dilution and amplification which are opposite to my understanding of these concepts

We hope our better explanation of these concepts in lines 28-36 and our responses to your earlier comments have helped to clear up these confusions. Additionally, we altered this sentence in the Discussion (lines 223-225) slightly to say that the lower tick densities in the center of the enclosure are caused specifically by the limitations of rodent-based tick dispersal, in order to be more specific.

Discussion is good on effects of rodents and exclosure size, but same comments as above on dilution/amplification and interpretation of results from tick movement model, and study not including results from all exclosure field studies.

Hopefully our responses to your earlier comments and the corresponding changes we made in the text have resolved these points of contention. To summarize, we have included more clear and general definitions of dilution and amplification in the Introduction (lines 28-36) that should make our use of these terms in the Discussion seem more appropriate; we are opting to keep discussing questing tick movement due to its importance to our model results and its existence being supported in the literature for some species (cited in lines 75, 121-122, and 178); and we are choosing not to do an exhaustive search of all exclosure field studies since our paper is not a review or meta-analysis and thus doing so would be beyond the scope of our paper.